

^{35}Na β^- decay (2.1 ms) 2013StZY,2022Cr03,1983La12

Parent: ^{35}Na : $E=0$; $J^\pi=(3/2^+)$; $T_{1/2}=2.1$ ms 4; $Q(\beta^-)=22190$ syst; $\% \beta^-$ decay=100

^{35}Na - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Na .

^{35}Na - $Q(\beta^-)$: 22190 720 (syst,2021Wa16).

^{35}Na - $\% \beta^-$ decay: $\% \beta^- = 100$, $\% \beta^- n > 0$.

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , $\beta\gamma$ -coin, and implant- β correlation, and deduced the $T_{1/2}$ of ^{35}Na . Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}\text{Ca}^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr_3 detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event by event by energy loss in the upstream PIN detector (Z determination) and time of flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- β correlation and deduced $T_{1/2}$ of ^{35}Na .

Comparisons with QRPA and shell-model calculations.

1983La12,1984La03: ^{35}Na was produced via the fragmentation of a 30 g/cm² iridium target by 10-GeV protons from the CERN synchrotron. The recoiling nuclear fragments are thermalized in heated graphite. The alkali elements selectively diffuse out of the target, are surface ionized, and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector. β particles were detected using a well-shaped thin plastic scintillator. Measured β and neutron counts. Deduced the $T_{1/2}$ of ^{35}Na . P_n values were deduced for other isotopes but not for ^{35}Na .

^{35}Na β^- -delayed neutrons have been observed by 1983La12. Experimental $\% \beta^- n$ values are unknown.

 $\gamma(^{35}\text{Mg})$

E_γ	Comments
^x 661	2013StZY observed this γ ray in the background-subtracted γ -ray spectrum in a time window of 0-10 ms after the implantation of ^{35}Na but did not observe it in later time windows. 2013StZY proposed that it could either be the deexcitation of the first 2^+ to 0^+ g.s. in ^{34}Mg or the deexcitation of an excited state in ^{35}Mg . Based on theoretical calculations (2019Mo01,2021Mi17), the $\beta^- 1n$ should be the dominant branch, implying that this γ ray is more likely from ^{34}Mg .

^x γ ray not placed in level scheme.